

Mentor Update — Evidence-Based Health Insurance Claim Adjudication Pilot

Project: M.Tech research — *Evidence-Based Health Insurance Claim Adjudication Under Policy and Clinical Ambiguity*
Date: 20 September 2026 | **Repository:** `D:\Claim_Adjudication_Pilot`

1. Progress summary

1.1 Foundation (complete)

- Frozen 24-case benchmark loaded and validated (`data/claim_adjudication_milestone1_benchmark.xlsx`).
- Every field classified as **inference / ground truth / metadata**; baselines may only consume `InferenceCase` via `prepare_inference_case()`, and ground truth/metadata are never exposed during inference (leakage-safe by construction).
- Config-driven evaluation framework (`configs/evaluation.yaml`, seed 42) with a shared evaluator and a fixed metric bundle.

1.2 Baselines (all implemented and evaluated)

ID	Baseline	Approach	Status
B1	Rule-based	Lexical rule engine: policy parsing (exclusion, EXCEPT, UNLESS, ONLY_IF), negation/uncertainty detection, synonym expansion, missing-evidence → HUMAN_REVIEW	Complete
B2	KAMEL-inspired	Concept extraction + biomedical semantic matcher (S-PubMedBert-MS-MARCO) + knowledge layer (is-a relations) + policy linker	Complete
B3	Single LLM/RAG	FAISS retriever over leak-safe policy corpus (top-k=5) + OpenCode LLM adjudicator	1/24 errored

1.3 Test suite

- 110 pytest tests, all passing** (loader / models / validator, schema, result validation, and per-baseline decision behaviours with offline stubs).

1.4 Verification trail

- Portable artifact `evaluation_result.json` confirms a sample single-case adjudication (glaucoma → DENY, confidence 0.98).

2. Benchmark results (24 cases; seed 42)

Metric	B1 rule-based	B2 KAMEL-inspired	B3 single LLM/RAG
Accuracy	1.0000	1.0000	0.9130
Macro precision	1.0000	1.0000	0.9259
Macro recall	1.0000	1.0000	0.9048
Macro F1	1.0000	1.0000	0.9028
APPROVE F1 (support 7)	1.0000	1.0000	0.8333

DENY F1 (support 10)	1.0000	1.0000	1.0000
HUMAN_REVIEW F1 (support 7)	1.0000	1.0000	0.8750
False denial rate	0.0000	0.0000	0.0000
False approval rate	0.0000	0.0000	0.0000
Human-review precision	1.0000	1.0000	0.7778
Average latency	≈3 ms	≈5 s	≈52 s
Cases evaluated / errored	24 / 0	24 / 0	23 / 1

Key observations

- B1 and B2 both reach perfect accuracy; B2 adds far more non-lexical coverage (semantic + hierarchy relations) at ≈5 s/case.
- B3 misclassified 2 APPROVE cases as HUMAN_REVIEW and errored on 1 case (LLM call failure). LLM confidence is a self-reported heuristic, not a calibrated probability.
- No false approvals or false denials in any baseline — errors are confined to over-conservative escalation.

3. Project folder structure

```

Claim_Adjudication_Pilot/
├── configs/
│   └── evaluation.yaml          # run config (seed, paths, thresholds, model names)
├── data/
│   └── claim_adjudication_milestone1_benchmark.xlsx  # frozen benchmark (24 cases)
├── src/
│   ├── common/
│   │   ├── constants.py        # column map, field roles, decision vocabulary
│   │   └── schemas.py          # InferenceCase, PredictionResult, prepare_inference_case()
│   ├── data/
│   │   ├── loader.py           # workbook -> AdjudicationCase
│   │   ├── models.py           # canonical case model + leakage-safe inference view
│   │   └── validator.py        # schema/domain checks -> DatasetReport
│   ├── baselines/
│   │   ├── rule_based/         # B1: rule engine + adjudicator + CLI (run.py)
│   │   ├── kamel_inspired/     # B2: concept_extractor, semantic_matcher,
│   │   │                       # medical_knowledge, policy_linker, adjudicator, CLI
│   │   └── llm_rag/            # B3: corpus + leakage guard, faiss retriever,
│   │                           # llm_client (opencode), adjudicator, CLI
│   └── evaluation/
│       ├── config.py           # EvaluationConfig + YAML loader
│       ├── evaluator.py        # run_evaluation, per-case timing, error capture
│       └── metrics.py          # fixed metric bundle (accuracy, macro, rates, latency)
├── results/
│   ├── metrics/                # JSON + CSV per baseline (stable + timestamped names)
│   ├── raw/                    # per-case predictions JSONL + retrieved-evidence log (B3)
│   └── reports/                 # markdown summaries + per-case prediction CSVs
├── tests/                      # 110 pytest tests (foundation + B1/B2/B3)
├── evaluation_result.json      # sample single-case adjudication artifact
├── pyproject.toml
├── requirements.txt
└── README.md

```

4. Next steps

1. **Investigate B3's single errored case** and harden LLM failure handling (retry/fallback so a failed call does not drop a case).
2. **Error analysis for B3:** 2 genuine APPROVEs routed to HUMAN_REVIEW; review prompt design and retrieval quality, and assess calibration of the self-reported confidence score.
3. **Sensitivity analysis:** vary KAMEL similarity threshold (currently 0.92), retrieval top-k, and prompt framing; break results down by `evidence_state` / `policy_logic_type` / `difficulty`.
4. **Implement the proposed evidence-reconciliation multi-agent system** — the primary research contribution — reusing the shared `InferenceCase` / `AdjudicationPrediction` contract and evaluator.
5. **Comparative evaluation** of B1 vs B2 vs B3 vs the proposed system, plus module-level (component) metrics on top of the frozen framework metric bundle.
6. **Benchmark expansion** beyond 24 cases to strengthen statistical confidence and domain coverage.

5. Open questions for mentor

- Should the multi-agent system (next step 4) be optimised for **accuracy ceiling** (outperform B1/B2) or for **robustness** (resolve ambiguities that force HUMAN_REVIEW today)?
- Is the comparative evaluation expected to be purely framework-level, or should module-level (component) metrics also be reported?
- Any preference on LLM fail-over policy for B3 (retry vs. mark-as-error) before we formalise the evaluation protocol?